

Problem Statement

Breast cancer is one of the most common cancers affecting women worldwide and remains a major cause of cancer-related deaths. Early detection is critical for improving treatment outcomes and increasing survival rates. However, accurately diagnosing breast cancer requires the analysis of multiple diagnostic features, making the process complex and time-consuming.

Delays in diagnosis or incorrect classification of tumors can affect timely medical intervention and patient care. Therefore, there is a growing need for an intelligent system that can assist healthcare professionals in analyzing diagnostic data and accurately classifying breast tumors as **Benign** or **Malignant**, enabling faster, more reliable, and data-driven clinical decision-making.